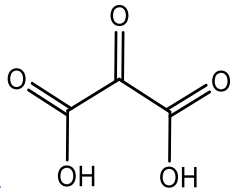
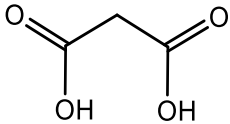
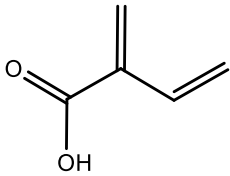
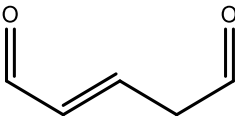
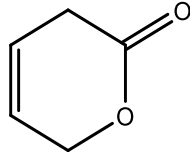


ACJC solutions for H2 Chemistry Prelim Paper 3 2022

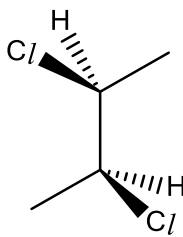
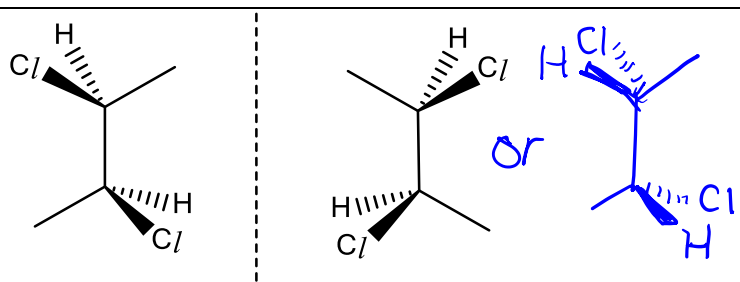
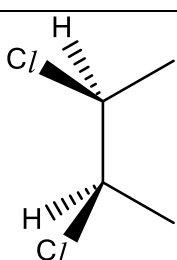
Section A

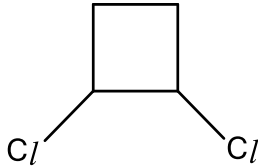
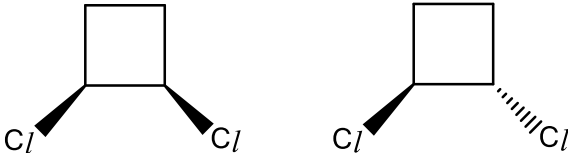
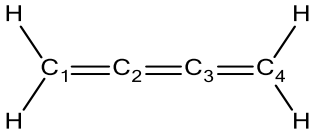
1	(a)	Explain what is meant by the term <i>Lewis acid</i> . [1]												
		Lewis acid is a substance that accepts an electron pair from a base via a dative covalent bond (i.e. an electron-pair acceptor)												
	(b)	The water molecule can react in various ways: as an acid, as a base, as a nucleophile, as an oxidising agent and as a reducing agent. Study the following reactions and decide in which way water is reacting in each case. Explain your answers fully. [5]												
	(i)	$2\text{F}_2 + 2\text{H}_2\text{O} \rightarrow 4\text{HF} + \text{O}_2$												
		Reducing agent as itself is being oxidised. The oxidation number of O increases from -2 in H_2O to 0 in O_2. Or water reduces F_2 to HF as the oxidation number of F decreases from 0 in F_2 to -1 in HF.												
	(ii)	$\text{CH}_3\text{COC}l + \text{H}_2\text{O} \rightarrow \text{CH}_3\text{COOH} + \text{HCl}$												
		Nucleophile. The carbon of the acyl chloride functional group is electron deficient and is susceptible to nucleophilic attack by water, which has a lone pair of electrons on its oxygen atom.												
	(iii)	$\text{H}_2\text{PO}_4^- + \text{H}_2\text{O} \rightleftharpoons \text{HPO}_4^{2-} + \text{H}_3\text{O}^+$												
		Base. H_2O accepts a proton from H_2PO_4^- to form H_3O^+ and HPO_4^{2-} .												
	(iv)	$\text{Li} + \text{H}_2\text{O} \rightarrow \text{LiOH} + \frac{1}{2} \text{H}_2$												
		Oxidising agent. Water oxidises Li to Li^+ as the oxidation number of Li increases from 0 in Li to +1 in LiOH												
	(v)	$\text{NO}_2^- + \text{H}_2\text{O} \rightleftharpoons \text{HNO}_2 + \text{OH}^-$												
		Acid. H_2O donates a proton to NO_2^- to form OH^- and HNO_2 .												
	(c)	The pK_a values of three acids are listed in the Table 1.1 below: Table 1.1 <table border="1"> <thead> <tr> <th>acid</th><th>formula</th><th>pK_a</th></tr> </thead> <tbody> <tr> <td>1</td><td>$\text{CH}_3\text{CH}_2\text{COOH}$</td><td>4.9</td></tr> <tr> <td>2</td><td>$\text{CH}_3\text{CHC}l\text{COOH}$</td><td>2.8</td></tr> <tr> <td>3</td><td>$\text{CH}_2\text{C}l\text{CH}_2\text{COOH}$</td><td>z</td></tr> </tbody> </table>	acid	formula	pK_a	1	$\text{CH}_3\text{CH}_2\text{COOH}$	4.9	2	$\text{CH}_3\text{CHC}l\text{COOH}$	2.8	3	$\text{CH}_2\text{C}l\text{CH}_2\text{COOH}$	z
acid	formula	pK_a												
1	$\text{CH}_3\text{CH}_2\text{COOH}$	4.9												
2	$\text{CH}_3\text{CHC}l\text{COOH}$	2.8												
3	$\text{CH}_2\text{C}l\text{CH}_2\text{COOH}$	z												
	(i)	Explain the difference in pK_a values between acid 1 and acid 2. [2]												

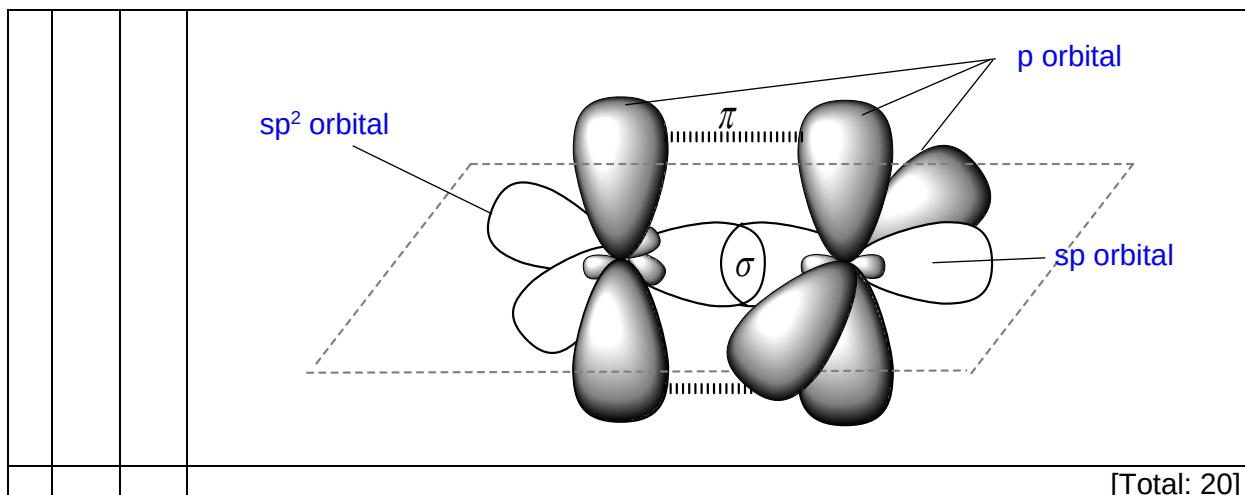
			 D is  E is  F is HOOC-COOH
		(ii)	Deduce the structures of A , B and C with reasoning. [7]
			A, B and C have 3 degrees of unsaturation. A, B and C have alkene functional group that undergoes electrophilic addition with bromine water. A, B and C undergoes vigorous oxidation / oxidative cleavage of the alkene functional group. A has the carboxylic acid functional group that undergoes acid-carbonate / acid-base reaction with carbonate. B has the aldehyde functional group that undergoes oxidation in the presence of Fehling's solution. C undergoes alkaline hydrolysis, so it has an ester functional group.  A  B  C
			[Total: 20]

2	(a)	The acid-base behaviour of aluminium oxide, Al_2O_3, shows similarities to that of magnesium oxide, MgO, on the one hand, and sulfur trioxide, SO_3, on the other. Describe what these similarities are, and explain why aluminium oxide occupies this in-between position. Write equations for all the reactions you choose to illustrate your answer. [5]
		MgO is an ionic oxide. Hence it is basic and is able to react with acids. $\text{MgO} + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{H}_2\text{O} \quad \text{or} \quad \text{MgO} + 2\text{H}^+ \rightarrow \text{Mg}^{2+} + \text{H}_2\text{O}$ SO₃ is a covalent oxide, hence it is acidic and is able to react with bases. $\text{SO}_3 + 2\text{NaOH} \rightarrow \text{Na}_2\text{SO}_4 + \text{H}_2\text{O} \quad \text{or} \quad \text{SO}_3 + 2\text{OH}^- \rightarrow \text{SO}_4^{2-} + \text{H}_2\text{O} \quad \text{or} \quad \text{SO}_3 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_4$ Al₂O₃: amphoteric oxide hence it react with both acids and bases. $\text{Al}_2\text{O}_3 + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + 3\text{H}_2\text{O} \quad \text{or} \quad \text{Al}_2\text{O}_3 + 6\text{H}^+ \rightarrow 2\text{Al}^{3+} + 3\text{H}_2\text{O}$ $\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{Na}[\text{Al}(\text{OH})_4]^- \quad \text{or} \quad \text{Al}_2\text{O}_3 + 2\text{OH}^- + 3\text{H}_2\text{O} \rightarrow 2[\text{Al}(\text{OH})_4]^-$ Al³⁺ has high charge density/polarizing power which is able to polarise the electron cloud of O^{2-}/anion, hence giving the oxide some covalent character.

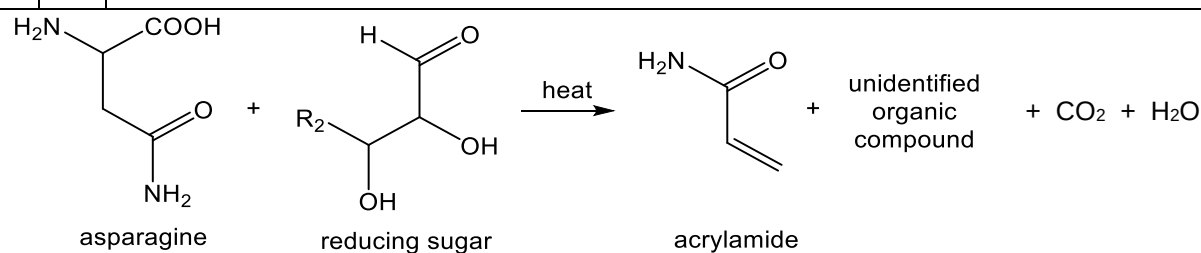
		Therefore, Al ₂ O ₃ has a giant ionic lattice structure with covalent character, making it an amphoteric oxide that reacts with both acids and alkalis.								
(b)	(i)	Explain what is meant by the term <i>standard enthalpy change of atomisation of magnesium</i>. [1] Standard enthalpy change of atomisation of magnesium is the energy absorbed/required/taken in when one mole of gaseous Mg atom/Mg(g) is formed from Mg at 298 K and 1 bar.								
	(ii)	An experiment was carried out by adding 0.05 g of magnesium ribbon to 50 cm³ of excess hydrochloric acid solution. The increase in temperature was 4.5 °C. The reaction is represented by the ionic equation as shown: $\text{Mg(s)} + 2\text{H}^+(\text{aq}) \rightarrow \text{Mg}^{2+}(\text{aq}) + \text{H}_2(\text{g}) \quad \Delta H_1^\ominus$ Use data from the <i>Data Booklet</i> to calculate $\Delta H_1^\ominus$, the standard enthalpy change of this reaction. [2] Heat released = $mc\Delta T$ = 50 x 4.18 x 4.5 = 940.5 J $\Delta H_1^\ominus = - [940.5 / (0.05 / 24.3)]$ = - 457 kJ mol⁻¹								
	(iii)	Besides using an experimental method, another way to determine $\Delta H_1^\ominus$, the standard enthalpy change of the above reaction, is to construct an energy cycle. Use the data in Table 2.1, together with data from the <i>Data Booklet</i>, to calculate another value of $\Delta H_1^\ominus$. [4] Table 2.1 <table><tr><td></td><td>$\Delta H^\ominus / \text{kJ mol}^{-1}$</td></tr><tr><td>standard enthalpy change of atomisation of Mg(s)</td><td>+148</td></tr><tr><td>standard enthalpy change of hydration of Mg²⁺(g)</td><td>-1921</td></tr><tr><td>standard enthalpy change of hydration of H⁺(g)</td><td>-1090</td></tr></table>		$\Delta H^\ominus / \text{kJ mol}^{-1}$	standard enthalpy change of atomisation of Mg(s)	+148	standard enthalpy change of hydration of Mg ²⁺ (g)	-1921	standard enthalpy change of hydration of H ⁺ (g)	-1090
	$\Delta H^\ominus / \text{kJ mol}^{-1}$									
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			$ \begin{array}{c} \text{Mg(s)} + 2\text{H}^+(\text{aq}) \xrightarrow{\Delta H_1^\ominus} \text{Mg}^{2+}(\text{aq}) + \text{H}_2(\text{g}) \\ \begin{array}{ccc} \downarrow \Delta H_{\text{atom}}^\ominus & & \downarrow -2 \times \Delta H_{\text{hyd}}^\ominus \text{ of } \text{H}^+(\text{g}) \\ \text{Mg(g)} + 2\text{H}^+(\text{g}) & & \\ \downarrow 1^{\text{st}} \text{ IE of Mg} & & \downarrow -2 \times 1^{\text{st}} \text{ IE of H(g)} \\ \text{Mg}^+(\text{g}) + 2\text{H(g)} & & \\ \downarrow 2^{\text{nd}} \text{ IE of Mg} & & \downarrow -\text{BE (H-H)} \\ \text{Mg}^{2+}(\text{g}) + \text{H}_2(\text{g}) & \xrightarrow{\Delta H_{\text{hyd}}^\ominus \text{ of Mg}^{2+}(\text{g})} & \end{array} \end{array} $ $\Delta H_1^\ominus = \Delta H_{\text{atom}}^\ominus + 1^{\text{st}} \text{ IE of Mg} + 2^{\text{nd}} \text{ IE of Mg} + (-2 \times \Delta H_{\text{hyd}}^\ominus \text{ of } \text{H}^+(\text{g})) + (-2 \times 1^{\text{st}} \text{ IE of H(g)}) + (-\text{BE (H-H)}) + \Delta H_{\text{hyd}}^\ominus \text{ of Mg}^{2+}(\text{g})$ $= +148 + 736 + 1450 + (-2 \times -1090) + (-2 \times 1310) + (-436) + (-1921)$ $= -463 \text{ kJ mol}^{-1} [1]$
	(c)	2,3-dichlorobutane exhibits stereoisomerism, one of which is enantiomerism. One enantiomer of 2,3-dichlorobutane is shown below.	
	(i)	Draw the other enantiomer.	[1]
			
	(ii)	A molecule with n chiral centres can have a maximum of 2^n stereoisomers. 2,3-dichlorobutane has two chiral centres but it only has three stereoisomers. Draw the third stereoisomer of 2,3-dichlorobutane and explain why it has only three stereoisomers.	[2]
			

			The diastereomer of 2,3-dichlorobutane is a meso compound which has an internal plane of symmetry. Hence, 2,3-dichlorobutane has a pair of enantiomers and only one diastereomer.
		(iii)	Suggest why 1,2-dichlorocyclobutane can exhibit cis-trans isomerism and draw the pair of cis-trans isomers. [2] 
			There is restricted rotation about the C–C bond that is bonded to the 2 Cl groups due to the presence of a ring structure. There are also two different groups bonded to each of the two adjacent carbon atoms with restricted rotation. Thus, 1,2-dichlorocyclobutane can exhibit cis-trans isomerism. 
		(iv)	A cumulene is a hydrocarbon with three or more consecutive double bonds. The simplest molecule in this class is butatriene, C₄H₄.  With the aid of a labelled diagram, describe the orbitals that form the C₁=C₂ bond in butatriene, and state the type of hybridisation involved. [3] In butatriene, the double bond of C=C is made of one σ bond and one π bond. C-C σ bond: Formed by head-on overlap of the sp² hybrid orbitals of C₁ atom and sp hybrid orbitals of C₂ atom. C-C π bond: Formed by sideways overlap of the two unhybridised p orbitals, forming an electron cloud above and below the plane.



3	(a)	The Maillard reaction is an important process that occurs between sugars with amino acids during cooking. Its occurrence is responsible for the golden brown colouration and crispy texture that makes food appealing and tasty. In 2002, it was discovered that Maillard reaction involving asparagine, an amino acid, forms acrylamide, a known carcinogen. Reaction 1 shows the chemical equation for the formation of acrylamide.
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reaction 1

		To determine the kinetics of reaction 1 , three experiments were carried out and the results are shown in Table 3.1.																			
		Table 3.1																			
		<table><tr><th rowspan="2">experiment</th><th colspan="2">concentration / mol dm⁻³</th><th rowspan="2">initial rate / 10⁻³ mol dm⁻³ s⁻¹</th></tr><tr><th>asparagine</th><th>reducing sugar</th></tr><tr><td>1</td><td>0.100</td><td>0.500</td><td>2.46</td></tr><tr><td>2</td><td>0.100</td><td>0.625</td><td>3.08</td></tr><tr><td>3</td><td>0.200</td><td>0.750</td><td>3.69</td></tr></table>		experiment	concentration / mol dm ⁻³		initial rate / 10 ⁻³ mol dm ⁻³ s ⁻¹	asparagine	reducing sugar	1	0.100	0.500	2.46	2	0.100	0.625	3.08	3	0.200	0.750	3.69
experiment	concentration / mol dm ⁻³		initial rate / 10 ⁻³ mol dm ⁻³ s ⁻¹																		
	asparagine	reducing sugar																			
1	0.100	0.500	2.46																		
2	0.100	0.625	3.08																		
3	0.200	0.750	3.69																		
		(i)	Deduce the order of reaction with respect to asparagine and the reducing sugar. Hence write the rate equation. [3]																		
			Comparing Experiment 1 and 2, When [reducing sugar] was increased 1.25x while keeping [asparagine] constant, rate of reaction was increased 1.25x.																		

		Order of reaction with respect to the reducing sugar is 1. Comparing Experiment 1 and 3, Let the rate equation be $\text{rate} = k[\text{reducing sugar}][\text{asparagine}]^m$ $\frac{\text{Rate 3}}{\text{Rate 1}} = \frac{k(0.750)(0.200)^m}{k(0.500)(0.100)^m}$ $\frac{0.00369}{0.00246} = \frac{k(0.750)(0.200)^m}{k(0.500)(0.100)^m}$ $m = 0$ Order of reaction with respect to asparagine is 0. OR Comparing Experiment 1 and 3, When [reducing sugar] x1.5 while keeping the rest constant, rate of reaction x1.5. Since observed rate of reaction was increased 1.5 times when [asparagine] was doubled, it implies that doubling [asparagine] has no effect on the rate. Order of reaction with respect to asparagine is 0. Rate = $k[\text{reducing sugar}]$
	(ii)	The reducing sugar is at an initial concentration of 1.60 mol dm^{-3} and its half-life is 140 s. Sketch a graph of concentration of the reducing sugar against time for reaction 1. [2]
	(iii)	Determine the rate constant, k, for reaction 1. [1] $t_{1/2} = \ln 2 \div k$ $k = \ln 2 \div t_{1/2} = \ln 2 \div 140 = 0.00495 \text{ s}^{-1}$ OR Using experiment 1,

		Rate = k [reducing sugar] $0.00246 = k \cdot 0.500$ $k = 0.00492 \text{ s}^{-1}$
	(b)	The reaction between the reducing sugar and aqueous sodium borohydride is a nucleophilic addition reaction. Draw the mechanism for this reaction, assuming that sodium borohydride produces the hydride ion, :H^-, as the reacting species. In your answer, you may represent the reducing sugar as RCHO. [3]
	(c)	The following mechanism in figure 3.1 has been proposed for the Maillard reaction involving asparagine.

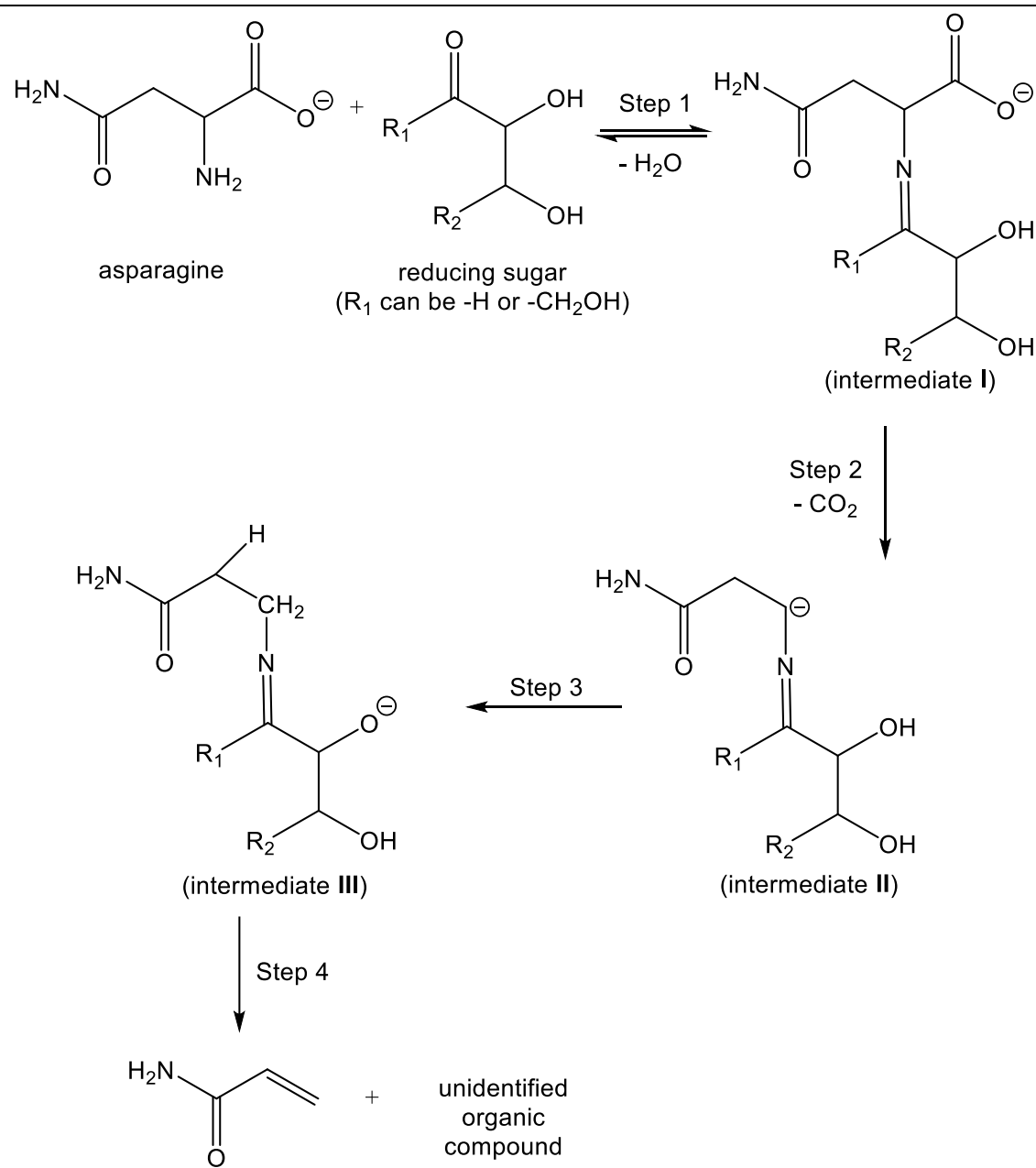


Fig. 3.1

	(i)	State the role of asparagine in step 1.	[1]
		Nucleophile	
	(ii)	Suggest and explain the effect on the rate of step 1 when R_1 is changed from $-H$ to $-CH_2OH$.	[1]
		The electron-donating alkyl (or $-CH_2OH$) group causes the carbonyl carbon to be less electron deficient. Hence there is a slower rate of nucleophilic addition. OR The $-CH_2OH$ group poses greater steric hindrance to the nucleophile, so there is a slower rate of nucleophilic addition. OR The electron-withdrawing $-CH_2OH$ group causes the carbonyl carbon to be more electron deficient. Hence there is a faster rate of nucleophilic addition.	

		(iii)	Draw the structure of the unidentified product of the above mechanism. [1]
		(d)	Compare and explain the base strength of the atoms N_a and N_b in asparagine. asparagine [2]
			N_b is less basic than N_a. The amide functional group at N_b is neutral / not basic. This is because the lone pair of electron on N_b delocalises to the electron withdrawing C=O group, making it unavailable to accept protons/H⁺ / coordinate to protons / protonation. OR The amine functional group at N_a is a stronger base. It is bonded to an electron-donating alkyl group that increases electron density at the N_a atom. This makes the lone pair of electrons on N_a more available to accept protons.
		(e)	In the 2020 Olympic Games, winners received medals made from recycled gold, silver and bronze. These metals were from recycled from electronic waste collected throughout Japan. In the recycling process, metal parts are physically removed from the plastics, and the metals are sorted by elements and levels of purity. Low purity silver is refined using electrolysis to remove metal impurities such as gold and copper.
		(i)	The electrode potential for Au⁺/Au is given below. $\text{Au}^+ + \text{e}^- \rightleftharpoons \text{Au} \quad E^\ominus = + 1.83 \text{ V}$ Define the term <i>standard electrode potential of Au⁺/Au</i>. [1]
			Standard Electrode (Redox) Potential, $E^\ominus$ is defined as the potential difference between a standard hydrogen electrode OR H⁺/H₂ half cell and the Au⁺/Au half cell under standard conditions at 1 mol dm⁻³, 25 °C / 298K and 1 bar.

		(ii)	Describe the electrode reactions that take place during this electrolysis and explain in detail how each of the two impurity metals is removed from the silver. [3]
			At the anode: $\text{Cu(s)} \rightarrow \text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \quad \text{and} \quad \text{Ag(s)} \rightarrow \text{Ag}^+(\text{aq}) + \text{e}^-$ Since the $E^\ominus_{\text{Cu}^{2+}/\text{Cu}}$ is more negative than $E^\ominus_{\text{Ag}^+/\text{Ag}}$, Cu is oxidised to Cu^{2+} ions at the anode, followed by Ag. Since the $E^\ominus_{\text{Au}^+/\text{Au}}$ is more positive than $E^\ominus_{\text{Ag}^+/\text{Ag}}$, Au will not be oxidised to Au^+ ions at the anode but will drop just off at the anode as anodic sludge. At the cathode: $\text{Ag}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{Ag(s)}$ Since $E^\ominus_{\text{Ag}^+/\text{Ag}}$ is more positive than $E^\ominus_{\text{Cu}^{2+}/\text{Cu}}$, Cu^{2+} will remain in the solution as Ag^+ is preferentially reduced over Cu^{2+}. Thus, both Cu and Au are removed from the silver.
		(iii)	The electrolytic refining of silver was carried out for 90 minutes using a current of 10.2 A. Using data from the <i>Data Booklet</i>, calculate the expected mass of silver deposited in this process. [2]
			$Q = (90 \times 60) \times 10.2 = 55080 \text{ C}$ $n(\text{e}) = 55080 \div 96500 = 0.5707 \text{ mol}$ $n(\text{Ag}) \text{ deposited} = 0.5707 \text{ mol}$ Mass of silver deposited = $0.5707 \times 107.9 = 61.6 \text{ g}$
			[Total: 20]

Section B

Answer **one** question from this section.

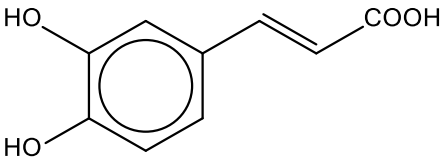
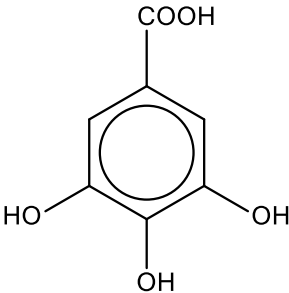
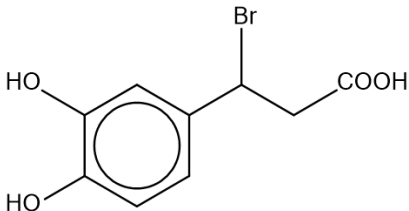
4	(a)	(i)	A glass rod was heated in a Bunsen burner flame and placed into a sample of hydrogen chloride gas. The experiment was repeated using a sample of hydrogen iodide gas. Use data from the <i>Data Booklet</i> to explain why a colour change was observed for hydrogen iodide but not hydrogen chloride. [2]
			The bond energy of H-I is 299 kJ mol^{-1} while bond energy of H-Cl is 431 kJ mol^{-1}. Less energy is required to break the weaker H-I bond than H-Cl, hence purple fumes of I_2 will be obtained for HI but no colour change for HCl [1] OR The heat energy in a Bunsen burner flame is sufficient to break H-I bond but not H-Cl bond. $2\text{HI (g)} \rightarrow \text{H}_2 \text{ (g)} + \text{I}_2 \text{ (g)}$

			A purple vapour I ₂ will be seen in the sample of hydrogen iodide gas.
		(ii)	Hydrogen chloride can be oxidised to chlorine and water in the presence of CuCl. Likewise, hydrogen cyanide can be oxidised to cyanogen and water in the presence of Cu according to the following equation. $2\text{HCN} + \frac{1}{2} \text{O}_2 \rightarrow (\text{CN})_2 + \text{H}_2\text{O}$ Describe the change in oxidation state of carbon which occurs during this reaction. [1]
			C in HCN is oxidised from +2 to +3 in (CN) ₂
		(iii)	Cyanogen can be converted to ethanedioic acid by heating with aqueous acid. State the type of reaction for this conversion. [1]
			Acid hydrolysis
			To determine the solubility product, K_{sp}, of calcium ethanedioate, CaC₂O₄, at 20 °C, an excess of solid calcium ethanedioate was shaken with 6.25 x 10⁻⁵ mol dm⁻³ of sodium ethanedioate. The resulting mixture was allowed to equilibrate at 20 °C in a water bath and then the mixture was filtered to obtain the filtrate, Solution A, which contains a saturated solution of calcium ethanedioate. 25.0 cm³ of Solution A was titrated with 5.0 × 10⁻⁵ mol dm⁻³ of acidified potassium manganate(VII). 16.60 cm³ of potassium manganate(VII) was required for complete reaction. The reaction of ethanedioate ions and acidified manganate(VII) ions is shown below. $2\text{MnO}_4^- + 5\text{C}_2\text{O}_4^{2-} + 16\text{H}^+ \rightarrow 2\text{Mn}^{2+} + 10\text{CO}_2 + 8\text{H}_2\text{O}$
	(b)	(i)	The reaction between ethanedioate ions and acidified potassium manganate(VII) has a high activation energy, hence solution A was heated to 60 °C before titration. Explain why the reaction has a high activation energy. [1]
			There is repulsion between two anions MnO ₄ ⁻ and C ₂ O ₄ ²⁻ thus energy is required to overcome the repulsion.
		(ii)	Calculate the amount of ethanedioate ions in 25.0 cm³ of Solution A. [2]
			Amount of MnO₄⁻ = $5.0 \times 10^{-5} \times \frac{16.60}{1000}$ = 8.30 x 10⁻⁷ mol Amount of ethanedioate ions in 25.0 cm³ of Solution A = $\frac{5}{2} \times 8.30 \times 10^{-7}$ = 2.08 x 10⁻⁶ mol
		(iii)	Calculate the concentration of ethanedioate ions in Solution A. [1]
			[ethanedioate ions] in 25.0 cm³ of Solution A = $\frac{1000}{25} \times 2.075 \times 10^{-6}$ = 8.30 x 10⁻⁵ mol dm⁻³

		(iv)	Calculate the concentration of calcium ions in solution A . [1]
			[ethanedioate ions] from $\text{CaC}_2\text{O}_4 = 8.30 \times 10^{-5} - 6.25 \times 10^{-5}$ $= 2.05 \times 10^{-5} \text{ mol dm}^{-3}$ $[\text{Ca}^{2+}] = 2.05 \times 10^{-5} \text{ mol dm}^{-3}$ since $[\text{Ca}^{2+}] = [\text{C}_2\text{O}_4^{2-}]$ from CaC_2O_4
		(v)	Write an expression for the solubility product, K_{sp} , of calcium ethanedioate and calculate its value at 20 °C, stating the units clearly in your answer. [3]
			$K_{\text{sp}} = [\text{Ca}^{2+}][\text{C}_2\text{O}_4^{2-}]$ $= (2.05 \times 10^{-5})(8.30 \times 10^{-5})$ $= 1.70 \times 10^{-9} \text{ mol}^2 \text{ dm}^{-6}$
	(c)		When a precipitate of calcium ethanedioate is formed, $\Delta G^\ominus_{\text{ppt}}$, in J mol^{-1}, is given by the following expression. $\Delta G^\ominus_{\text{ppt}} = RT \ln K_{\text{sp}}$
		(i)	Given that the K_{sp} value of calcium ethanedioate at 25 °C is $2.70 \times 10^{-9} \text{ mol}^2 \text{ dm}^{-6}$, calculate the $\Delta G^\ominus_{\text{ppt}}$ of calcium ethanedioate. [1]
			$\Delta G^\ominus_{\text{ppt}} = RT \ln K_{\text{sp}}$ $= 8.31 \times 298 \times \ln(2.70 \times 10^{-9})$ $= -48859 \text{ J mol}^{-1}$ $= -48.9 \text{ kJ mol}^{-1}$
		(ii)	The standard entropy change of the precipitation of calcium ethanedioate, $\Delta S^\ominus_{\text{ppt}}$ is $+93.2 \text{ J mol}^{-1} \text{ K}^{-1}$. Calculate the standard enthalpy change of precipitation of calcium ethanedioate, $\Delta H^\ominus_{\text{ppt}}$. [1]
			$-48859 = \Delta H^\ominus_{\text{ppt}} - 298(+93.2)$ $\Delta H^\ominus_{\text{ppt}} = -21.1 \text{ kJ mol}^{-1}$
	(d)		The precipitation of calcium ethanedioate is one of the reasons for the formation of kidney stones. Given that the stomach has a pH of about 2.0 while the kidney has a pH of about 6.5, explain why stones are not formed in the stomach but stones can be formed in the kidney. [2]
			At pH 2.0 in the stomach, $[\text{H}^+]$ is higher than at pH 6.5 in the kidney. The high $[\text{H}^+]$ in the stomach reacts with $\text{C}_2\text{O}_4^{2-}$ to form $\text{H}_2\text{C}_2\text{O}_4$ (or HC_2O_4^-), hence $[\text{C}_2\text{O}_4^{2-}]$ is much lower in the stomach than kidney. Thus the ionic product of calcium ethanedioate remains below K_{sp} in the stomach but can rise above K_{sp} in the kidney.

	(e)	Ethylenediaminetetraacetate, also known as EDTA has the structure shown below. Suggest a reason why EDTA can be used as a treatment for kidney stones. [2]
		EDTA can form a complex / chelates/ binds via a dative bond with Ca^{2+}, thus $[\text{Ca}^{2+}]$ decreases. Ionic product in the kidney can fall below K_{sp} thus the calcium ethanedioate precipitate will not form. (or is soluble) OR $\text{CaC}_2\text{O}_4(\text{s}) \rightleftharpoons \text{Ca}^{2+}(\text{aq}) + \text{C}_2\text{O}_4^{2-}(\text{aq})$ OR make reference to earlier part's equation. POE shifts right as $[\text{Ca}^{2+}]$ decreases, thus calcium ethanedioate dissolves.
	(f)	Calcium ethanedioate decomposes at a temperature of 400°C according to the equation shown below. $\text{CaC}_2\text{O}_4(\text{s}) \rightarrow \text{CaO}(\text{s}) + \text{CO}_2(\text{g}) + \text{CO}(\text{g})$ Explain why magnesium ethanedioate decomposes at a lower temperature than calcium ethanedioate. [2]
		Mg^{2+} has a higher charge density/polarising power than Ca^{2+}. The electron cloud of $\text{C}_2\text{O}_4^{2-}$ in MgC_2O_4 is distorted to a greater extent hence the bonds in anion are weakened more than in CaC_2O_4. Less energy is required to break the bonds in MgC_2O_4.
		[Total: 20]

5	(a)	The equilibrium constant, K_p, of the following reaction is 1.64 at a certain temperature. equation 1 $\text{I}_2(\text{s}) + \text{Br}_2(\text{g}) \rightleftharpoons 2\text{IBr}(\text{g}) \quad \Delta H < 0$ In a study, some bromine gas was introduced into a vessel with excess solid iodine at the same temperature and the reaction was allowed to take place until equilibrium was reached. The partial pressure of $\text{IBr}(\text{g})$ at equilibrium was found to be 0.80 atm. Assume that all the bromine is in the gaseous state and that the vapour pressure of iodine is negligible.																				
	(i)	State the K_p expression for equation 1. Hence, calculate the partial pressure of Br_2 at equilibrium. [2]																				
		$K_p = \frac{(P_{\text{IBr}})^2}{(P_{\text{Br}_2})}$ $1.64 = \frac{0.8^2}{(P_{\text{Br}_2})}$ $P_{\text{Br}_2} = \frac{0.8^2}{1.64} = 0.39024 = 0.390 \text{ atm}$																				
	(ii)	Hence calculate the initial pressure of Br_2 in the container. [1]																				
		<table><tr><td></td><td>$\text{I}_2(\text{s})$</td><td>+ $\text{Br}_2(\text{g})$</td><td>$\rightleftharpoons$</td><td>$2\text{IBr}(\text{g})$</td></tr><tr><td>Initial / atm</td><td>-</td><td>$x + 0.4$</td><td></td><td>0</td></tr><tr><td>Change / atm</td><td>-</td><td>-0.4</td><td></td><td>+0.80</td></tr><tr><td>Equilibrium / atm</td><td>-</td><td>x</td><td></td><td>0.80</td></tr></table> Initial pressure = $0.390 + 0.4 = 0.790 \text{ atm}$		$\text{I}_2(\text{s})$	+ $\text{Br}_2(\text{g})$	$\rightleftharpoons$	$2\text{IBr}(\text{g})$	Initial / atm	-	$x + 0.4$		0	Change / atm	-	-0.4		+0.80	Equilibrium / atm	-	x		0.80
	$\text{I}_2(\text{s})$	+ $\text{Br}_2(\text{g})$	$\rightleftharpoons$	$2\text{IBr}(\text{g})$																		
Initial / atm	-	$x + 0.4$		0																		
Change / atm	-	-0.4		+0.80																		
Equilibrium / atm	-	x		0.80																		
	(iii)	Sketch a graph of rate against time, showing how the rates of the forward and reverse reactions change with time, from the point of mixing to the point after equilibrium is established. Label your graphs clearly. [2]																				
	(iv)	State and explain the effect on the rate of production and yield of IBr when temperature is decreased. [3]																				
		As temperature is decreased, the system would favour the forward, exothermic reaction to release heat. POE shifts to the right. Frequency of effective collision decreases as fewer particles have kinetic energy greater than activation energy. The yield of IBr increases and rate of production of IBr decreases.																				

		(v)	State and explain the effect on the equilibrium partial pressure of Br ₂ when the pressure of the system is halved at constant temperature. [2]
			When the pressure of the system is halved, the system will respond by increasing the number of gaseous particles present, to increase pressure. This favours the forward reaction for the position of equilibrium to shift to the right and the partial pressure of Br₂ will decrease.
	b		Caffeic acid and gallic acid are organic compounds that can be found in coffee and tea leaves respectively.  caffeic acid  gallic acid
		(i)	Describe a chemical test to distinguish the above two compounds and state the expected observations. [2]
			Add KMnO₄, H₂SO₄(aq) to both compounds and heat For caffeic acid, purple KMnO₄ will decolorise. Colourless, odourless gas (CO₂) produced will form white ppt with limewater. For gallic acid, purple KMnO₄ will remain. Or add KMnO₄ in NaOH For caffeic acid, purple KMnO₄ will decolorise and brown ppt form while for gallic acid, purple KMnO₄ will remain. Or KMnO₄, H₂SO₄(aq) For caffeic acid, purple KMnO₄ will decolorise while for gallic acid, purple KMnO₄ will remain.
		(ii)	Draw the structure of the major product that is formed when caffeic acid reacts with HBr. [1]
			

	(c)	Buffer solutions are very important in many biological reactions as enzymes need a specific pH range to function properly. One of the most important buffer systems is the carbonic acid, H_2CO_3, and bicarbonate, HCO_3^-, buffer system in blood. $\text{H}_2\text{CO}_3(\text{aq}) \rightleftharpoons \text{H}^+(\text{aq}) + \text{HCO}_3^-(\text{aq})$ The above equilibrium has a K_a value of $7.90 \times 10^{-7} \text{ mol dm}^{-3}$ and is responsible for buffering the pH of the blood at 7.4.
	(i)	A blood buffer can be simulated by mixing $0.500 \text{ mol dm}^{-3} \text{ H}_2\text{CO}_3$ and 2.00 mol dm^{-3} of NaHCO_3. Show that the volumes of H_2CO_3 and NaHCO_3 required to make 200 cm^3 of this buffer solution with pH 7.4 are 34 cm^3 and 166 cm^3 respectively [3]
		$\text{p}K_a = -\lg (7.90 \times 10^{-7}) = 6.102$ $\text{pH} = \text{p}K_a + \lg \frac{[\text{HCO}_3^-]}{[\text{H}_2\text{CO}_3]}$ $7.4 = -\lg (7.90 \times 10^{-7}) + \lg \frac{[\text{HCO}_3^-]}{[\text{H}_2\text{CO}_3]}$ $\lg \frac{[\text{HCO}_3^-]}{[\text{H}_2\text{CO}_3]} = 1.298$ $\frac{[\text{HCO}_3^-]}{[\text{H}_2\text{CO}_3]} = 19.86$ Since volume of $\text{H}_2\text{CO}_3 + \text{HCO}_3^- = 0.200 \text{ dm}^3$, let the volume of HCO_3^- be $A \text{ dm}^3$ and the volume of H_2CO_3 be $(0.200 - A) \text{ dm}^3$. $\frac{\frac{A \times 2}{0.200}}{\frac{(0.200 - A) \times 0.5}{0.200}} = 19.86$ $A = 0.166 \text{ dm}^3$ $= 166 \text{ cm}^3$ Volume of $\text{HCO}_3^- = 166 \text{ cm}^3$ Volume of $\text{H}_2\text{CO}_3 = 200 - 166 = 34 \text{ cm}^3$ for showing both volumes
	(ii)	With the aid of an equation, explain how the buffer solution prepared in c(i) helps to maintain the pH at around 7.4 when a small amount of sodium hydroxide is added. [2]
		$\text{H}_2\text{CO}_3 + \text{OH}^- \rightarrow \text{HCO}_3^- + \text{H}_2\text{O}$ H_2CO_3 will react with hydroxide added to minimise the decrease in $[\text{H}^+]$/no significant increase in $[\text{OH}^-]$. Hence, the increase in pH of the solution is not significant.

		(iii)	Hence calculate the resultant pH of the above buffer solution when 0.003 mol of solid sodium hydroxide is added. [2]
			Amount of H_2CO_3 consumed = $[(34/1000) \times 0.5] - 0.003 = 0.0140 \text{ mol}$ Amount of HCO_3^- produced = $[(166/1000) \times 2] + 0.003 = 0.335 \text{ mol}$ $\text{pH} = \text{pK}_a + \lg \frac{[\text{HCO}_3^-]}{[\text{H}_2\text{CO}_3]} = 6.102 + \lg \frac{\frac{0.335}{0.200}}{\frac{0.200}{0.0140}} = 7.48$
			[Total: 20]